DUE: FRIDAY, SEPTEMBER 18, 2026

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Unless specified otherwise, for all questions in this HW, assume temperature = 300K (kT = 0.026 eV) and $n_i = 1.5 \times 10^{10}$ cm^{-3}.

1. CARRIER CONCENTRATION

(A). Show that the electron and hole distribution functions:

$$n_0 = N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right) \tag{1.1}$$

$$p_0 = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right) \tag{1.2}$$

are equivalent to

$$n_0 = n_i \exp\left(\frac{E_F - E_i}{k_B T}\right) \tag{1.3}$$

$$p_0 = n_i \exp\left(\frac{E_i - E_F}{k_B T}\right) \tag{1.4}$$

where E_i is the intrinsic energy level.

(B). Derive that

$$n_0 p_0 = n_i^2 \tag{1.5}$$

When is this equation valid?

Solution: 20 points total.

(A). **10 points.**
Start from the Maxwell–Boltzmann (MB) approximations for electrons and holes:

$$n_0 = N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right) \tag{1.6}$$

$$p_0 = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right) \tag{1.7}$$

The intrinsic carrier concentration written relative to the intrinsic level E_i is

$$n_i = N_C \exp\left(-\frac{E_C - E_i}{k_B T}\right) \tag{1.8}$$

$$n_i = N_V \exp\left(-\frac{E_i - E_V}{k_B T}\right) \tag{1.9}$$

Substitute (1.8) into (1.6):

$$n_0 = \left[N_C \exp\left(-\frac{E_C - E_i}{k_B T}\right)\right] \exp\left(\frac{E_F - E_i}{k_B T}\right) \tag{1.10}$$

which simplifies to

$$n_0 = n_i \exp\left(\frac{E_F - E_i}{k_B T}\right) \tag{1.11}$$

Similarly, substitute (1.9) into (1.7):

$$p_0 = \left[N_V \exp\left(-\frac{E_i - E_V}{k_B T}\right)\right] \exp\left(\frac{E_i - E_F}{k_B T}\right) \tag{1.12}$$

which gives

$$p_0 = n_i \exp\left(\frac{E_i - E_F}{k_B T}\right) \tag{1.13}$$

(B). **10 points.**

Multiply (1.11) and (1.13):

$$n_0 p_0 = \left[n_i \exp\left(\frac{E_F - E_i}{k_B T}\right)\right]\left[n_i \exp\left(\frac{E_i - E_F}{k_B T}\right)\right] = n_i^2 \tag{1.14}$$

This result is known as the *mass-action law*. It is valid when the Maxwell–Boltzmann approximation holds, which requires that the Fermi level lie sufficiently far from the band edges (typically more than ~ $3k_B T$ away from E_C and E_V). Under this condition, carrier statistics are nondegenerate and the exponential MB expressions for carrier concentrations are accurate.

2. TEMPERATURE EFFECTS

Consider a piece of silicon doped with acceptors $N_A = 1.0 \times 10^{17}\ \text{cm}^{-3}$. Assume complete ionization. At $T = 300$ K, the intrinsic carrier concentration is $n_i = 1.5 \times 10^{10}\ \text{cm}^{-3}$, and at $T = 500$ K, $n_i = 8.4 \times 10^{13}\ \text{cm}^{-3}$. Use $k_B = 8.617 \times 10^{-5}$ eV/K.

(A). At room temperature, compare the mobilities of two Si samples doped with $N_D = 10^{15}\ \text{cm}^{-3}$ and $N_D = 10^{19}\ \text{cm}^{-3}$. Which one has the higher mobility, and why?

(B). The Fermi level relative to the intrinsic level is

$$E_F - E_i = -k_B T \ln\left(\frac{N_A}{n_i}\right) \tag{2.1}$$

Calculate $E_F - E_i$ at $T = 300$ K and $T = 500$ K. Comment on how the Fermi level shifts with temperature.

(C). Using charge neutrality together with eq. (1.5), calculate the equilibrium electron concentration n_0 in this sample at $T = 300$ K and $T = 500$ K. Comment on whether the semiconductor is extrinsic or intrinsic at each temperature.

Solution: 20 points total.

(A). **6 points.**

At $N_D = 10^{15}\ \text{cm}^{-3}$ (lightly doped), carrier mobility is relatively high. At $N_D = 10^{19}\ \text{cm}^{-3}$(heavily doped), the density of ionized impurities is much larger, leading tO increased impurity scattering and a much lower mobility.

Comment: Carrier mobility decreases as dopant concentration increases due to enhanced ionized impurity scattering.

(B). **6 points.**

Using

$$E_F - E_i = -k_B T \ln\left(\frac{N_A}{n_i}\right)$$

with $N_A = 1.0 \times 10^{17}\ \text{cm}^{-3}$:

At $T = 300$ K:

$$\ln\left(\frac{N_A}{n_i}\right) = \ln\left(\frac{10^{17}}{1.5\times10^{10}}\right) \approx 15.71 \tag{2.2}$$

$$E_F - E_i = -(8.617\times10^{-5})(300)(15.71) \approx -0.41\ \text{eV} \tag{2.3}$$

At $T = 500$ K:

$$\ln\left(\frac{N_A}{n_i}\right) = \ln\left(\frac{10^{17}}{8.4\times10^{13}}\right) \approx 7.08 \tag{2.4}$$

$$E_F - E_i = -(8.617\times10^{-5})(500)(7.08) \approx -0.31\ \text{eV} \tag{2.5}$$

Comment: As temperature increases, the intrinsic carrier concentration n_i increases, reducing the ratio N_A/n_i. As a result, the magnitude of $E_F - E_i$ decreases, meaning the Fermi level moves closer to the intrinsic level.

(C). **8 points.**

Charge neutrality and mass-action law give:

$$p_0 + N_D = n_0 + N_A, \qquad n_0 p_0 = n_i^2$$

With $N_D = 0$, eliminate p_0:

$$n_0(n_0 + N_A) = n_i^2 \tag{2.6}$$

$$n_0 = \frac{-N_A + \sqrt{N_A^2 + 4n_i^2}}{2} \tag{2.7}$$

At $T = 300$ K ($n_i = 1.5\times10^{10}\ \text{cm}^{-3}$):

$$n_0 \approx \frac{-1.0\times10^{17} + \sqrt{(1.0\times10^{17})^2 + 4(1.5\times10^{10})^2}}{2} \tag{2.8}$$

$$\approx 2.25\times10^{3}\ \text{cm}^{-3} \tag{2.9}$$

At $T = 500$ K ($n_i = 8.4\times10^{13}\ \text{cm}^{-3}$):

$$n_0 \approx \frac{-1.0\times10^{17} + \sqrt{(1.0\times10^{17})^2 + 4(8.4\times10^{13})^2}}{2} \tag{2.10}$$

$$\approx 7.06\times10^{10}\ \text{cm}^{-3} \tag{2.11}$$

Comment: At 300 K, $p_0 \approx N_A \gg n_i$ and $p_0 \gg n_0$, so the semiconductor is clearly extrinsic. Even at 500 K, $p_0 \approx N_A \gg n_i$, so the material remains extrinsic, although intrinsic carriers become more significant. Only at much higher temperatures would n_i become comparable to N_A and intrinsic behavior dominate.

3. BAND DIAGRAM

A piece of Si uniformly doped with acceptors at a concentration of $N_A = 5 \times 10^{13}\ \text{cm}^{-3}$ is at equilibrium at $T = 300$ K. Assume complete ionization of dopants. The intrinsic carrier concentration is $n_i = 1.5 \times 10^{10}\ \text{cm}^{-3}$.

(A). Write the charge neutrality equation for this semiconductor and use it to determine the equilibrium hole and electron concentrations.

(B). Where is the Fermi level E_F relative to the intrinsic level E_i? Calculate $E_i - E_F$.

(C). Draw the equilibrium energy band diagram for this silicon. The horizontal axis is position, and the vertical axis is energy. Clearly label E_F, E_i, E_C, and E_V and indicate their relative positions.

Solution: 20 points total.

(A). **6 points.**

Start from the full charge neutrality condition:

$$p_0 - n_0 + N_D^+ - N_A^- = 0 \tag{3.1}$$

For this sample, the silicon is uniformly doped only with acceptors, so

$$N_D = 0 \ \Rightarrow \ N_D^+ \approx 0 \tag{3.2}$$

Assuming complete ionization of acceptors,

$$N_A^- \approx N_A \tag{3.3}$$

The charge neutrality equation reduces to:

$$p_0 - n_0 - N_A = 0 \tag{3.4}$$

Since the material is p-type and $N_A \gg n_i$, the hole concentration is much larger than the electron concentration, so $p_0 \gg n_0$. Thus,

$$p_0 \approx N_A = 5 \times 10^{13}\ \text{cm}^{-3} \tag{3.5}$$

The electron concentration is then obtained from the mass-action law:

$$n_0 = \frac{n_i^2}{p_0} = \frac{(1.5 \times 10^{10})^2}{5 \times 10^{13}} = 4.5 \times 10^{6}\ \text{cm}^{-3} \tag{3.6}$$

(B). **6 points.**

The Fermi level position relative to the intrinsic level is given by:

$$E_i - E_F = k_B T \ln\left(\frac{p_0}{n_i}\right) \tag{3.7}$$

Using the approximation from Part (A),

$$p_0 \approx N_A \tag{3.8}$$

$$E_i - E_F = k_B T \ln\left(\frac{N_A}{n_i}\right) \tag{3.9}$$

$$= (0.0259\ \text{eV}) \ln\left(\frac{5 \times 10^{13}}{1.5 \times 10^{10}}\right) \tag{3.10}$$

$$= 0.21\ \text{eV} \tag{3.11}$$

Thus, the Fermi level lies below the intrinsic energy level.

(C). **8 points.**

Since the material is p-type, the Fermi level lies closer to the valence band. The equilibrium energy band diagram is shown in Figure 3.1, with E_F, E_i, E_C, and E_V clearly labeled.

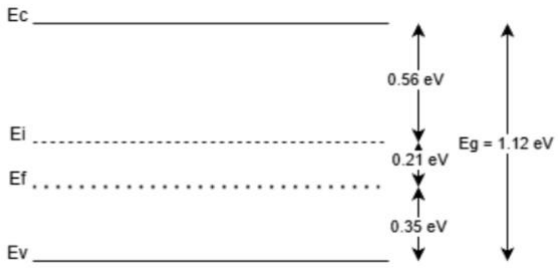


Figure 3.1: Band Diagram

4. RESISTANCE UNDER DOPING

A Si bar has a length of 10 mm and a cross-sectional area of 1 mm^2. At $T = 300$ K, the intrinsic carrier concentration is $n_i = 1.5 \times 10^{10}\ cm^{-3}$. Mobility values can be read from the provided figure. Calculate its resistance in the following cases:

(A). Without any doping.

(B). Doped with $5 \times 10^{17}\ cm^{-3}$ phosphorus.

(C). Doped with $5 \times 10^{17}\ cm^{-3}$ phosphorus and $2 \times 10^{16}\ cm^{-3}$ boron.

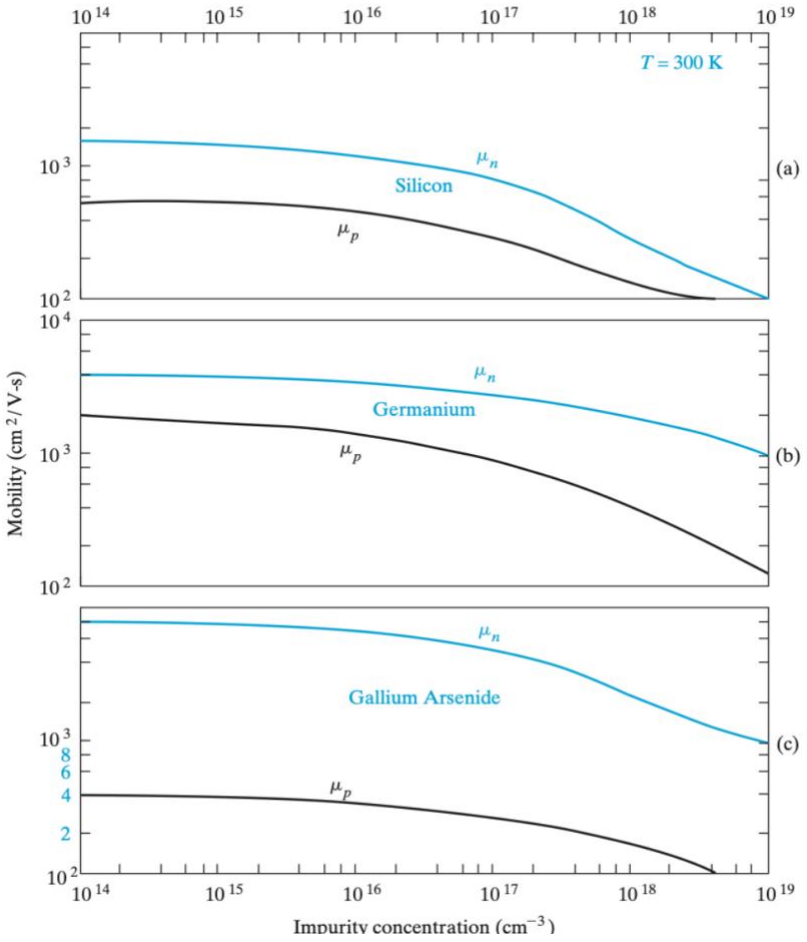


Figure 3–23: Variation of mobility with total doping impurity concentration $(N_a + N_d)$ for Ge, Si, and GaAs at 300 K.

Solution: 20 points total.

(A). **6 points.**

For $N_D = N_A = 0$:

$$\mu_n = 1350\ \text{cm}^2/\text{V}\cdot\text{s}, \qquad \mu_p = 450\ \text{cm}^2/\text{V}\cdot\text{s} \tag{4.1}$$

$$\sigma = qn_0\mu_n + qp_0\mu_p = q\left(n_i\mu_n + n_i\mu_p\right) = 4.32\times10^{-4}\ (\Omega\cdot m)^{-1} \tag{4.2}$$

$$R = \frac{L}{\sigma A} = 23\ \text{M}\Omega \tag{4.3}$$

(B). **6 points.**

For $N_D = 5\times10^{17}\ \text{cm}^{-3}$:

$$\mu_n = 380\ \text{cm}^2/\text{V}\cdot\text{s} \tag{4.4}$$

$$\sigma = qn_0\mu_n + qp_0\mu_p = qN_D\mu_n = 3.04\times10^{3}\ (\Omega\cdot m)^{-1} \tag{4.5}$$

$$R = \frac{L}{\sigma A} = 3.3\ \Omega \tag{4.6}$$

(C). **8 points.**

For $N_D = 5\times10^{17}\ \text{cm}^{-3}$, $N_A = 2\times10^{16}\ \text{cm}^{-3}$:

$$N_D + N_A = 5.2\times10^{17}\ \text{cm}^{-3}, \qquad N_D - N_A = 4.8\times10^{17}\ \text{cm}^{-3} \Rightarrow \mu_n = 370\ \text{cm}^2/\text{V}\cdot\text{s} \tag{4.7}$$

$$\sigma = qn_0\mu_n + qp_0\mu_p = q(N_D - N_A)\mu_n = 2.84\times10^{3}\ (\Omega\cdot m)^{-1} \tag{4.8}$$

$$R = \frac{L}{\sigma A} = 3.5\ \Omega \tag{4.9}$$